COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 6

8 Logic and Proof (mj201)

Resolution, unification

Which of the following formulas is valid? Prove them using resolution or find a counterexample. If a counterexample exists, explain how resolution is blocked from producing a spurious proof.

(*a*) $(\forall x\,[\neg Q(x,x) \to P(x)]) \to \forall x\,[P(x) \lor \exists y\, Q(x,y)]$ [7 marks]

Answer: Refute and convert to CNF:

$$\begin{array}{rl}
\text{Refute:} & \neg[(\forall x\,[\neg Q(x,x) \to P(x)]) \to \forall x\,(P(x) \lor \exists y\, Q(x,y))] \\
\text{convert:} & (\forall x\,[\neg Q(x,x) \to P(x)]) \land \neg\forall x\,(P(x) \lor \exists y\, Q(x,y)) \\
& (\forall x\,[Q(x,x) \lor P(x)]) \land \exists x\,\neg(P(x) \lor \exists y\, Q(x,y)) \\
& (\forall x\,[Q(x,x) \lor P(x)]) \land \exists x\,(\neg P(x) \land \forall y\,\neg Q(x,y)) \\
\text{Skolemise:} & (\forall x\,[Q(x,x) \lor P(x)]) \land (\neg P(a) \land \forall y\,\neg Q(a,y)) \\
\text{clauses:} & Q(x,x) \lor P(x),\ \neg P(a),\ \neg Q(a,y)
\end{array}$$

Then, resolution (with unification):

$$\dfrac{\dfrac{Q(x,x) \lor P(x) \quad \neg P(a)}{Q(a,a)} \qquad \neg Q(a,y)}{\square}$$

Thus we can conclude that $(\forall x\,(\neg Q(x,x) \to P(x))) \to \forall x\,(P(x) \lor \exists y\, Q(x,y))$ is valid.

(*b*) $[\exists x\, P(x)] \to \exists x\, P(f(x))$ [6 marks]

Answer: The statement is not valid! What happens if we try to prove it by resolution?

$$\begin{array}{rl}
\text{Refute:} & \neg([\exists x\, P(x)] \to \exists x\, P(f(x))) \\
\text{convert:} & [\exists x\, P(x)] \land \forall x\,\neg P(f(x)) \\
\text{Skolemise:} & P(a) \land \forall x\,\neg P(f(x)) \\
\text{clauses:} & P(a),\ \neg P(f(x))
\end{array}$$

But we cannot do resolution because a and $f(x)$ cannot be unified! We can use this to inform a counterexample. We just need to find an interpretation for P and f where P holds for some elements, but not for any of the form $f(x)$. For example, in arithmetic, if $f(x) = 2x$ and P is the property of being odd then $\exists x\, \texttt{odd}(x)$ is true but $\exists x\, \texttt{odd}(2x)$ is false, demonstrating that $(\exists x\, P(x)) \to \exists x\, P(f(x))$ is not valid.

(*c*) $\exists z\, \forall w\, [(\forall x\, [P(x) \to \exists y\, Q(x,y)]) \to (P(w) \to Q(w,z))]$ [7 marks]

Answer: The statement is not valid! What happens if we try to prove it by resolution?

Refute and convert to CNF:

$$\begin{array}{rl}
\text{Refute:} & \neg\exists z\, \forall w\, [(\forall x\, [P(x) \to \exists y\, Q(x,y)]) \to (P(w) \to Q(w,z))] \\
\text{convert:} & \forall z\, \exists w\, \neg[(\forall x\, [P(x) \to \exists y\, Q(x,y)] \to (P(w) \to Q(w,z)))] \\
& \forall z\, \exists w\, [\forall x\, [\neg P(x) \lor \exists y\, Q(x,y)] \land P(w) \land \neg Q(w,z)] \\
\text{Skolemise:} & \forall z\, [\forall x\, [\neg P(x) \lor Q(x, g(x))] \land P(f(z)) \land \neg Q(f(z), z)] \\
\text{clauses:} & \neg P(x) \lor Q(x, g(x)),\ P(f(z)),\ \text{and } \neg Q(f(z), z)
\end{array}$$

If we try to prove it by resolution we get an occurs check fail:

$$\dfrac{\dfrac{\dfrac{\neg P(x) \lor Q(x, g(x)) \quad P(f(z))}{Q(f(z), g(f(z)))} \qquad \neg Q(f(z), z)}{\text{occurs check fail}}}{\square}$$

Counterexample: $P(x)$ can be a trivial property (satisfied by any x) and $Q(x, y)$ can be the property $x = y$.